THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 9 — MATHS FULL SYLLABUS PAPER SET 06

Full Syllabus (Chapters 1–8) | RBSE Pattern | Session 2026–27

Time: 3 Hours 15 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 80**

General Instructions:

1. All questions are compulsory.
2. Q. 1–20 are Objective Type (1 mark each).
3. Q. 21–28 are Very Short Answer Type (2 marks each).
4. Q. 29–36 are Short Answer Type (3 marks each).
5. Q. 37–40 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 35 to 40.
7. Write the question number before each answer.
8. Draw neat figures wherever required.

SECTION — A (Objective Type Questions) 20 × 1 = 20

Choose the correct option and write it in your answer book.

1. The distance between $A(1, 2)$ and $B(4, 6)$ is: 1

(a) 5 units **(b)** 4 units **(c)** 7 units **(d)** 6 units

2. The reflection of the point $(-2, 6)$ in the y-axis is: 1

(a) $(6, -2)$ **(b)** $(-2, -6)$ **(c)** $(2, -6)$ **(d)** $(2, 6)$

3. $M(0, 0)$ is the midpoint of AB and $A(-3, 7)$. The coordinates of B are: 1

(a) $(3, -7)$ **(b)** $(3, 7)$ **(c)** $(-3, 7)$ **(d)** $(4, -6)$

4. The sequence $7, 12, 17, 22, \ldots$ forms a linear pattern. Its nth term is: 1

(a) $5n + 2$ **(b)** $5n + 7$ **(c)** $5n - 12$ **(d)** $7n + 5$

5. The value of $2x^2 - 1x + 1$ at $x = -1$ is: 1

(a) 1 **(b)** 4 **(c)** 2 **(d)** 6

6. The number $0.1\overline{6}$ in the form $\frac{p}{q}$ is: 1

(a) $\frac{1}{6}$ **(b)** $\frac{1}{3}$ **(c)** 6 **(d)** $\frac{4}{15}$

7. The number $0.333\ldots$ is: 1

(a) irrational **(b)** rational

(c) neither rational nor irrational **(d)** not a real number

8. Rational numbers are NOT closed under: 1

(a) addition **(b)** multiplication

(c) subtraction **(d)** division by zero (undefined)

9. The side of a cube of volume $p^3 + 6p^2q + 12pq^2 + 8q^3$ is: 1

(a) $p + 2q$ **(b)** $p + 4q$ **(c)** $2p + q$ **(d)** $p + q$

10. $(x + 7)(x + 8)$ equals: 1

(a) $x^2 + 56x + 15$ **(b)** $x^2 - x + 56$ **(c)** $x^2 + 15x - 56$ **(d)** $x^2 + 15x + 56$

11. In a circle of radius 13 cm, a chord is 12 cm from the centre. The length of the chord is: 1

(a) 8 cm **(b)** 10 cm **(c)** 12 cm **(d)** 5 cm

12. ABCD is a cyclic quadrilateral with $\angle A = 123^\circ$. Then $\angle C$ is: 1

(a) 237° **(b)** 29° **(c)** 123° **(d)** 57°

13. Through two given points the number of circles that can be drawn is: 1

(a) exactly one **(b)** infinitely many **(c)** exactly two **(d)** none

14. By Heron's formula, the area of a triangle with sides 10 cm, 17 cm and 21 cm is: 1

(a) 170 cm^2 **(b)** 90 cm^2 **(c)** 84 cm^2 **(d)** 78 cm^2

15. Taking $\pi = \frac{22}{7}$, the circumference of a circle of radius 3.5 cm is: 1

(a) 11 cm **(b)** 44 cm **(c)** 22 cm **(d)** 38.5 cm

16. The area of a parallelogram with base 20 cm and height 9 cm is: 1

(a) 90 cm^2 **(b)** 58 cm^2 **(c)** 200 cm^2 **(d)** 180 cm^2

17. A fair die is rolled once. The probability of getting a multiple of 3 is: 1

(a) $\frac{1}{2}$ **(b)** $\frac{1}{4}$ **(c)** $\frac{2}{3}$ **(d)** $\frac{1}{3}$

18. An event occurred 46 times in 200 trials. Its experimental probability (relative frequency) is: 1

(a) 0.23 **(b)** 4.35 **(c)** 0.115 **(d)** 0.33

19. Which term of the sequence $t_n = 2n - 1$ is 137? 1

(a) $n = 70$ **(b)** $n = 68$ **(c)** $n = 71$ **(d)** $n = 69$

20. The 10th term of the AP $3, 8, 13, \ldots$ is: 1

(a) 49 **(b)** 96 **(c)** 48 **(d)** 47

SECTION — B (Very Short Answer Type) — 8 × 2 = 16

Answer in about 20–30 words each.

21. The points $A(2,\ 0)$, $B(0,\ 2)$ and $C(-2,\ 0)$ are joined to form a triangle. Name the type of triangle with reason. 2

22. A chess club charges ₹200 joining fee plus ₹50 per match. Write the expression for the amount paid for m matches and find m if a player paid ₹750. 2

23. Show that $0.\overline{9} = 1$ using algebra. 2

24. Factorise $x^2 - 5x + 6$ by splitting the middle term. 2

25. In a circle of radius 12 cm, the chord AB subtends 60° at the centre O. Find AB. 2

26. Why does an inscribed regular hexagon show that $\pi > 3$? 2

27. A card is drawn from a well-shuffled pack of 52 cards. Find the probability that it is 2
(i) a king
(ii) a red king.

28. Find the nth term of the AP $11, 8, 5, 2, \ldots$ and write its recursive rule. 2

SECTION — C (Short Answer Type) — 8 × 3 = 24

Answer in about 40–60 words each.

29. Triangle ADM has vertices $A(3, 4)$, $D(7, 1)$, $M(9, 6)$. It is reflected in the y-axis. Write the image vertices and show that $A'D' = AD$. 3

30. A student has ₹500 in her account and receives ₹150 every month. Write a linear expression for the amount in the nth month (from the second month onwards) and find in which month she will have ₹1,850. 3

31. Explain, using a debt example, why $10 - (-5) = 15$. 3

32. Show that $(a+b)^2 + (a-b)^2 = 2(a^2+b^2)$ and use it to find $(105)^2 + (95)^2$. 3

33. A circular ring has outer radius 14 cm and inner radius 7 cm. Find its area ($\pi = \frac{22}{7}$). 3

34. A coin was tossed 1000 times with 520 heads. Find the relative frequency of heads and the difference from the theoretical probability. 3

35. ABCD is a cyclic trapezium with $AB \parallel CD$. Show that its non-parallel sides AD and BC are equal. 3

OR

Two chords AB (30 cm) and CD (16 cm) lie on the same side of the centre of a circle of radius 17 cm. Find the distance between them. 3

36. In the Sierpiński triangle the black area at Stage 0 is 1 sq. unit. Find the black area at Stage 3 and show it equals (number of black triangles) × (area of each). 3

OR

Derive the formula for the sum of the first n natural numbers. 3

SECTION — D (Long Answer Type) — 4 × 5 = 20

Answer in about 100–150 words each.

37. Case study: A tailor has a roll of 15 m of silk; each kurta needs 2 m. Another roll has $12\frac{1}{2}$ m. 5

(i) How many kurtas from the 15 m roll and how much silk is left?

(ii) How many from the second roll?

(iii) Express the silk left in each case as a rational number and find the total silk left.

OR

(i) Express $0.\overline{27}$ and $1.2\overline{5}$ in the form $\frac{p}{q}$. 5

(ii) Write $0.\overline{9}$ as a fraction and comment.

38. Prove that if the opposite angles of a quadrilateral ABCD add up to 180°, then it is cyclic (do the case where C lies outside the circle through A, B, D). 5

OR

ABCD is a cyclic quadrilateral with $\angle A = (3x+5)^\circ$, $\angle B = (2y+10)^\circ$, $\angle C = (5x-25)^\circ$, $\angle D = (4y-10)^\circ$. 5

(i) Find x and y.

(ii) Find all four angles.

(iii) Verify that the angle sum is 360°.

39. Case study: A car tyre has diameter 56 cm. ($\pi = \frac{22}{7}$) 5

(i) How far does the car move in one revolution?

(ii) How many revolutions does the tyre make in 1.76 km?

(iii) If the tyre turns 500 times per minute, find the speed in km/h.

(iv) A tyre of diameter 70 cm turns the same number of times; compare the distances covered.

OR

(i) Use Heron's formula to find the area of the triangle with sides 17 cm, 25 cm and 26 cm. 5

(ii) Find the lengths of its three altitudes.

(iii) Which altitude is the longest?

40. An AP has 50 terms with $t_3 = 12$ and $t_{50} = 106$. **5**

(i) Find a and d.

(ii) Find the 29th term.

(iii) Find the sum of all 50 terms.

OR

Case study: A child arranges marbles in rows of 1, 2, 3, … **5**

(i) How many marbles are in the first 10 rows?

(ii) How many in 25 rows?

(iii) How many rows are needed for exactly 210 marbles?

(iv) Which triangular number is the answer to (i)?

END OF QUESTION PAPER **Total: 80 Marks**

MASTERMIND